



10660 - Unlocking the power of the Near-Infrared: Detailed Measurement of the Stellar Initial Mass Function in Quiescent Galaxies at Intermediate-z

Cycle: 5, Proposal Category: GO

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Dr. Elham Eftekhari (PI) (ESA Member)	Universiteit Leiden
Prof. Mariska Kriek (CoI) (ESA Member) (CoPI)	Universiteit Leiden
Martje Slob (CoI) (ESA Member)	Universiteit Leiden
Chloe M. Cheng (CoI) (ESA Member)	Universiteit Leiden
Mr. Wout Goesaert (CoI) (ESA Member)	Universiteit Leiden
Dr. Ignacio Martin-Navarro (CoI) (ESA Member)	Instituto de Astrofisica de Canarias
Dr. Anna Ferre-Mateu (CoI) (ESA Member)	Instituto de Astrofisica de Canarias
Prof. Arjen van der Wel (CoI) (ESA Member)	Universiteit Gent
Dr. Francesco La Barbera (CoI) (ESA Member)	Osservatorio Astronomico di Capodimonte (OAC)
Aliza G. Beverage (CoI)	Carnegie Institution of Washington
Prof. Pieter van Dokkum (CoI) (US Admin CoI)	Yale University
Prof. Charlie Conroy (CoI)	Harvard University

OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	7	imf_dither3_13targs_o pen_nir2-filler (Obs 6)	NIRSpec MultiObject Spectroscopy	(1) IMF_apl_catalog_v4

ABSTRACT

JWST Proposal 10660 (Created: Thursday, April 30, 2026, 6:00:20PM Eastern Standard Time) - Overview

One of the most remarkable discoveries of JWST is the presence of massive quiescent galaxies already in place at $z > 7$ –10, far earlier than predicted by Λ -CDM models. Interpreting the stellar masses of these systems critically depends on assumptions about the stellar initial mass function (IMF), yet the IMF remains poorly constrained beyond the local Universe. Despite compelling evidence for IMF variations in the centers of nearby massive early-type galaxies (ETGs), the origin and evolution of these variations remain unknown.

This program will use JWST/NIRSpec to obtain deep rest-frame near-infrared (NIR) spectra of nine massive quiescent galaxies at $z \sim 0.7$ —the critical intermediate-redshift regime linking local early-type galaxies (ETGs) to the massive quiescent systems already in place at $z > 7$. Building upon existing ultra-deep optical spectroscopy, our observations will target IMF-sensitive NIR absorption features (FeH, NaI, MgI, AlI) to break the long-standing IMF–abundance degeneracy and constrain the detailed low-mass end shape of the IMF in individual galaxies.

By constraining the IMF shape at $z \sim 0.7$, we will directly test whether the excess of low-mass stars inferred in local massive galaxies was already established when these systems first quenched, or if the IMF evolved subsequently. Anchoring IMF constraints at this epoch with the same direct diagnostics used locally will provide the missing bridge between nearby IMF studies and the unexpectedly massive quiescent galaxies revealed by JWST at cosmic dawn.

OBSERVING DESCRIPTION

We propose to obtain deep rest-frame near-infrared (NIR) spectroscopy of massive quiescent galaxies at $z \sim 0.7$ using NIRSpec-MSA with the G235M/F170LP disperser/filter combination. These observations will provide medium-resolution ($R \sim 1000$) spectra covering key IMF-sensitive features such as FeH_{0.99}, NaI_{1.14}, MgI_{1.18}, and AlI_{1.31}. Quiescent galaxies from the LEGA-C survey form the primary sample and star-forming systems at higher redshifts are used as fillers. All targets have high-quality HST/F814W imaging and are covered by the COSMOS-Web program (Cycle 1); therefore, no pre-imaging is required.

We will use 3-shutter slits and a custom 2-position nodding pattern, extending slits to 4–5 shutters where possible to maximize throughput and improve background subtraction. We apply a fixed dither of one shutter in the dispersion direction to mitigate detector defects and ensure full wavelength sampling.

This configuration consists of 36 dithers, executed in two visits. Each exposure uses the NRSIRS2 readout mode with 20 groups per integration and 2 integrations per exposure. The total charged time is 37.9 hours. Our field can be observed for a limited PA range (64–85° and 234–254°). The optimal mask has a PA of 247°, targeting 9 primary galaxies; due to the high density of sources, most allowable PAs will still include at least 8

primaries. All observations will therefore adopt a PA of $\sim 247^\circ$ to maintain the optimized MSA design and maximize primary target coverage within the COSMOS field.

Proposal 10660 - Targets - Unlocking the power of the Near-Infrared: Detailed Measurement of the Stellar Initial Mass Function in Qui...

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	IMF_apl_catalog_v4	RA: 09 59 41.1705 (149.9215437d) Dec: +02 29 46.44 (2.49623d) Equinox: J2000		
	Comments: Description=[]				

Proposal 10660 - Observation 7 - Unlocking the power of the Near-Infrared: Detailed Measurement of the Stellar Initial Mass Function ...

Observation	Proposal 10660, Observation 7: imf_dither3_13targs_open_nir2-filler (Obs 6) Diagnostic Status: Warning Observing Template: NIRSpec MultiObject Spectroscopy <i>Comments: The Aperture PA Range has been approved by TTRB (TTRBJWST-89574) in order to be the same as in our earlier Program 5629 observations.</i>
Diagnostics	(imf_dither3_13targs_open_nir2-filler (Obs 6) (Obs 7)) Warning (Form): Config c1 (#1) has 1 filler slits affected by failed closed shutters. (imf_dither3_13targs_open_nir2-filler (Obs 6) (Obs 7)) Warning (Form): Config c1 (#10) has 1 filler slits affected by failed closed shutters. (imf_dither3_13targs_open_nir2-filler (Obs 6) (Obs 7)) Warning (Form): Config c1 (#11) has 1 filler slits affected by failed closed shutters. (imf_dither3_13targs_open_nir2-filler (Obs 6) (Obs 7)) Warning (Form): Config c1 (#12) has 1 filler slits affected by failed closed shutters. (imf_dither3_13targs_open_nir2-filler (Obs 6) (Obs 7)) Warning (Form): Config c1 (#13) has 1 filler slits affected by failed closed shutters. (imf_dither3_13targs_open_nir2-filler (Obs 6) (Obs 7)) Warning (Form): Config c1 (#14) has 1 filler slits affected by failed closed shutters. (imf_dither3_13targs_open_nir2-filler (Obs 6) (Obs 7)) Warning (Form): Config c1 (#15) has 1 filler slits affected by failed closed shutters. (imf_dither3_13targs_open_nir2-filler (Obs 6) (Obs 7)) Warning (Form): Config c1 (#16) has 1 filler slits affected by failed closed shutters. (imf_dither3_13targs_open_nir2-filler (Obs 6) (Obs 7)) Warning (Form): Config c1 (#17) has 1 filler slits affected by failed closed shutters. (imf_dither3_13targs_open_nir2-filler (Obs 6) (Obs 7)) Warning (Form): Config c1 (#18) has 1 filler slits affected by failed closed shutters. (imf_dither3_13targs_open_nir2-filler (Obs 6) (Obs 7)) Warning (Form): Config c1 (#2) has 1 filler slits affected by failed closed shutters. (imf_dither3_13targs_open_nir2-filler (Obs 6) (Obs 7)) Warning (Form): Config c1 (#3) has 1 filler slits affected by failed closed shutters. (imf_dither3_13targs_open_nir2-filler (Obs 6) (Obs 7)) Warning (Form): Config c1 (#4) has 1 filler slits affected by failed closed shutters. (imf_dither3_13targs_open_nir2-filler (Obs 6) (Obs 7)) Warning (Form): Config c1 (#5) has 1 filler slits affected by failed closed shutters. (imf_dither3_13targs_open_nir2-filler (Obs 6) (Obs 7)) Warning (Form): Config c1 (#6) has 1 filler slits affected by failed closed shutters. (imf_dither3_13targs_open_nir2-filler (Obs 6) (Obs 7)) Warning (Form): Config c1 (#7) has 1 filler slits affected by failed closed shutters. (imf_dither3_13targs_open_nir2-filler (Obs 6) (Obs 7)) Warning (Form): Config c1 (#8) has 1 filler slits affected by failed closed shutters. (imf_dither3_13targs_open_nir2-filler (Obs 6) (Obs 7)) Warning (Form): Config c1 (#9) has 1 filler slits affected by failed closed shutters. (imf_dither3_13targs_open_nir2-filler (Obs 6) (Obs 7)) Warning (Form): Config c2 (#19) has 1 master background shutters affected by failed open or closed shutters. (imf_dither3_13targs_open_nir2-filler (Obs 6) (Obs 7)) Warning (Form): Config c2 (#20) has 1 filler slits affected by failed closed shutters. (imf_dither3_13targs_open_nir2-filler (Obs 6) (Obs 7)) Warning (Form): Config c2 (#21) has 1 filler slits affected by failed closed shutters. (imf_dither3_13targs_open_nir2-filler (Obs 6) (Obs 7)) Warning (Form): Config c2 (#22) has 1 master background shutters affected by failed open or closed shutters. (imf_dither3_13targs_open_nir2-filler (Obs 6) (Obs 7)) Warning (Form): Config c2 (#23) has 1 master background shutters affected by failed open or closed shutters. (imf_dither3_13targs_open_nir2-filler (Obs 6) (Obs 7)) Warning (Form): Config c2 (#24) has 1 filler slits affected by failed closed shutters. (imf_dither3_13targs_open_nir2-filler (Obs 6) (Obs 7)) Warning (Form): Config c2 (#25) has 1 filler slits affected by failed closed shutters. (imf_dither3_13targs_open_nir2-filler (Obs 6) (Obs 7)) Warning (Form): Config c2 (#26) has 1 master background shutters affected by failed open or closed shutters. (imf_dither3_13targs_open_nir2-filler (Obs 6) (Obs 7)) Warning (Form): Config c2 (#27) has 1 master background shutters affected by failed open or closed shutters. (imf_dither3_13targs_open_nir2-filler (Obs 6) (Obs 7)) Warning (Form): Config c2 (#28) has 1 filler slits affected by failed closed shutters. (imf_dither3_13targs_open_nir2-filler (Obs 6) (Obs 7)) Warning (Form): Config c2 (#29) has 1 filler slits affected by failed closed shutters. (imf_dither3_13targs_open_nir2-filler (Obs 6) (Obs 7)) Warning (Form): Config c2 (#30) has 1 master background shutters affected by failed open or closed shutters. (imf_dither3_13targs_open_nir2-filler (Obs 6) (Obs 7)) Warning (Form): Config c2 (#31) has 1 master background shutters affected by failed open or closed shutters. (imf_dither3_13targs_open_nir2-filler (Obs 6) (Obs 7)) Warning (Form): Config c2 (#32) has 1 filler slits affected by failed closed shutters. (imf_dither3_13targs_open_nir2-filler (Obs 6) (Obs 7)) Warning (Form): Config c2 (#33) has 1 filler slits affected by failed closed shutters. (imf_dither3_13targs_open_nir2-filler (Obs 6) (Obs 7)) Warning (Form): Config c2 (#34) has 1 master background shutters affected by failed open or closed shutters. (imf_dither3_13targs_open_nir2-filler (Obs 6) (Obs 7)) Warning (Form): Config c2 (#35) has 1 master background shutters affected by failed open or closed shutters. (imf_dither3_13targs_open_nir2-filler (Obs 6) (Obs 7)) Warning (Form): Config c2 (#36) has 1 filler slits affected by failed closed shutters. (Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. (Visit 7:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Proposal 10660 - Observation 7 - Unlocking the power of the Near-Infrared: Detailed Measurement of the Stellar Initial Mass Function ...

Fixed Targets											
	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	IMF_apt_catalog_v4	RA: 09 59 41.1705 (149.9215437d) Dec: +02 29 46.44 (2.49623d) Equinox: J2000								
	Comments: Description=[]										
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
	2		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
Template	TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map		Spectral Overlap Threshold		
	MSATA	false	No	MSA Center	imf (34 sources)	filler_nir (2024 sources)	jwst-nirspec-g235m		1.5		
Reference Stars											

Proposal 10660 - Observation 7 - Unlocking the power of the Near-Infrared: Detailed Measurement of the Stellar Initial Mass Function ...

Spectral Elements	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (G235M/F170LP)	c1		149.91775416666 664 Degrees 2.50515 Degrees	246.80711421909 814		1.0	1	2	2946.956
	2	1 (G235M/F170LP)	c1		149.91775416666 664 Degrees 2.50515 Degrees	246.80712563415 84		-1.0	1	2	2946.956
	3	1 (G235M/F170LP)	c1		149.91775416666 664 Degrees 2.50515 Degrees	246.80712563415 84		-1.0	1	2	2946.956
	4	1 (G235M/F170LP)	c1		149.91775416666 664 Degrees 2.50515 Degrees	246.80711421909 814		1.0	1	2	2946.956
	5	1 (G235M/F170LP)	c1		149.91775416666 664 Degrees 2.50515 Degrees	246.80711421909 814		1.0	1	2	2946.956
	6	1 (G235M/F170LP)	c1		149.91775416666 664 Degrees 2.50515 Degrees	246.80712563415 84		-1.0	1	2	2946.956
	7	1 (G235M/F170LP)	c1		149.91775416666 664 Degrees 2.50515 Degrees	246.80712563415 84		-1.0	1	2	2946.956
	8	1 (G235M/F170LP)	c1		149.91775416666 664 Degrees 2.50515 Degrees	246.80711421909 814		1.0	1	2	2946.956
	9	1 (G235M/F170LP)	c1		149.91775416666 664 Degrees 2.50515 Degrees	246.80711421909 814		1.0	1	2	2946.956
	10	1 (G235M/F170LP)	c1		149.91775416666 664 Degrees 2.50515 Degrees	246.80712563415 84		-1.0	1	2	2946.956
	11	1 (G235M/F170LP)	c1		149.91775416666 664 Degrees 2.50515 Degrees	246.80712563415 84		-1.0	1	2	2946.956
	12	1 (G235M/F170LP)	c1		149.91775416666 664 Degrees 2.50515 Degrees	246.80711421909 814		1.0	1	2	2946.956
	13	1 (G235M/F170LP)	c1		149.91775416666 664 Degrees 2.50515 Degrees	246.80711421909 814		1.0	1	2	2946.956
	14	1 (G235M/F170LP)	c1		149.91775416666 664 Degrees 2.50515 Degrees	246.80712563415 84		-1.0	1	2	2946.956
	15	1 (G235M/F170LP)	c1		149.91775416666 664 Degrees 2.50515 Degrees	246.80712563415 84		-1.0	1	2	2946.956
	16	1 (G235M/F170LP)	c1		149.91775416666 664 Degrees 2.50515 Degrees	246.80711421909 814		1.0	1	2	2946.956
	17	1 (G235M/F170LP)	c1		149.91775416666 664 Degrees 2.50515 Degrees	246.80711421909 814		1.0	1	2	2946.956
	18	1 (G235M/F170LP)	c1		149.91775416666 664 Degrees 2.50515 Degrees	246.80712563415 84		-1.0	1	2	2946.956

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#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
19	1 (G235M/F170LP)	c2		149.91784195833 33 Degrees 2.504945 Degrees	246.80712979969 653		-1.0	1	2	2946.956
20	1 (G235M/F170LP)	c2		149.91784195833 33 Degrees 2.504945 Degrees	246.80711838560 427		1.0	1	2	2946.956
21	1 (G235M/F170LP)	c2		149.91784195833 33 Degrees 2.504945 Degrees	246.80711838560 427		1.0	1	2	2946.956
22	1 (G235M/F170LP)	c2		149.91784195833 33 Degrees 2.504945 Degrees	246.80712979969 653		-1.0	1	2	2946.956
23	1 (G235M/F170LP)	c2		149.91784195833 33 Degrees 2.504945 Degrees	246.80712979969 653		-1.0	1	2	2946.956
24	1 (G235M/F170LP)	c2		149.91784195833 33 Degrees 2.504945 Degrees	246.80711838560 427		1.0	1	2	2946.956
25	1 (G235M/F170LP)	c2		149.91784195833 33 Degrees 2.504945 Degrees	246.80711838560 427		1.0	1	2	2946.956
26	1 (G235M/F170LP)	c2		149.91784195833 33 Degrees 2.504945 Degrees	246.80712979969 653		-1.0	1	2	2946.956
27	1 (G235M/F170LP)	c2		149.91784195833 33 Degrees 2.504945 Degrees	246.80712979969 653		-1.0	1	2	2946.956
28	1 (G235M/F170LP)	c2		149.91784195833 33 Degrees 2.504945 Degrees	246.80711838560 427		1.0	1	2	2946.956
29	1 (G235M/F170LP)	c2		149.91784195833 33 Degrees 2.504945 Degrees	246.80711838560 427		1.0	1	2	2946.956
30	1 (G235M/F170LP)	c2		149.91784195833 33 Degrees 2.504945 Degrees	246.80712979969 653		-1.0	1	2	2946.956
31	1 (G235M/F170LP)	c2		149.91784195833 33 Degrees 2.504945 Degrees	246.80712979969 653		-1.0	1	2	2946.956
32	1 (G235M/F170LP)	c2		149.91784195833 33 Degrees 2.504945 Degrees	246.80711838560 427		1.0	1	2	2946.956
33	1 (G235M/F170LP)	c2		149.91784195833 33 Degrees 2.504945 Degrees	246.80711838560 427		1.0	1	2	2946.956
34	1 (G235M/F170LP)	c2		149.91784195833 33 Degrees 2.504945 Degrees	246.80712979969 653		-1.0	1	2	2946.956
35	1 (G235M/F170LP)	c2		149.91784195833 33 Degrees 2.504945 Degrees	246.80712979969 653		-1.0	1	2	2946.956
36	1 (G235M/F170LP)	c2		149.91784195833 33 Degrees 2.504945 Degrees	246.80711838560 427		1.0	1	2	2946.956

Special Requirements	Group Visits within 53.0 Days Aperture PA Range 246.8073 to 246.8073 Degrees (V3 108.2327303 to 108.2327303) Visits Same PA MSA Planned Aperture PA 246.8073 to 246.8073 Degrees (V3 108.2327303 to 108.2327303)
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